

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error
from sklearn.metrics import mean_absolute_error
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
%matplotlib inline
```

```
In [2]: df = pd.read_csv("BostonHousing.csv")
df = df.dropna()
df
```

```
Out[2]:
```

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO
0	0.00632	18.0	2.31	0.0	0.538	6.575	65.2	4.0900	1	296	15.3
1	0.02731	0.0	7.07	0.0	0.469	6.421	78.9	4.9671	2	242	17.8
2	0.02729	0.0	7.07	0.0	0.469	7.185	61.1	4.9671	2	242	17.8
3	0.03237	0.0	2.18	0.0	0.458	6.998	45.8	6.0622	3	222	18.7
5	0.02985	0.0	2.18	0.0	0.458	6.430	58.7	6.0622	3	222	18.7
...	...	...	...	...	...	...	...	...	...	...	...
499	0.17783	0.0	9.69	0.0	0.585	5.569	73.5	2.3999	6	391	19.2
500	0.22438	0.0	9.69	0.0	0.585	6.027	79.7	2.4982	6	391	19.2
502	0.04527	0.0	11.93	0.0	0.573	6.120	76.7	2.2875	1	273	21.0
503	0.06076	0.0	11.93	0.0	0.573	6.976	91.0	2.1675	1	273	21.0
504	0.10959	0.0	11.93	0.0	0.573	6.794	89.3	2.3889	1	273	21.0

394 rows × 14 columns

```
In [3]: df.head()
```

```
Out[3]:
```

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO
0	0.00632	18.0	2.31	0.0	0.538	6.575	65.2	4.0900	1	296	15.3
1	0.02731	0.0	7.07	0.0	0.469	6.421	78.9	4.9671	2	242	17.8
2	0.02729	0.0	7.07	0.0	0.469	7.185	61.1	4.9671	2	242	17.8
3	0.03237	0.0	2.18	0.0	0.458	6.998	45.8	6.0622	3	222	18.7
5	0.02985	0.0	2.18	0.0	0.458	6.430	58.7	6.0622	3	222	18.7

```
In [4]: df.columns
```

```
Out[4]: Index(['CRIM', 'ZN', 'INDUS', 'CHAS', 'NOX', 'RM', 'AGE', 'DIS', 'RAD',  
            'TAX',  
            'PTRATIO', 'B', 'LSTAT', 'MEDV'],  
            dtype='object')
```

```
In [10]: x = df[['CRIM', 'ZN', 'INDUS', 'CHAS', 'NOX', 'RM', 'AGE',  
                'DIS', 'RAD', 'TAX', 'PTRATIO', 'B', 'LSTAT']]  
x.head()
```

```
Out[10]:
```

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	
0	0.00632	18.0	2.31	0.0	0.538	6.575	65.2	4.0900	1	296	15.3	394
1	0.02731	0.0	7.07	0.0	0.469	6.421	78.9	4.9671	2	242	17.8	394
2	0.02729	0.0	7.07	0.0	0.469	7.185	61.1	4.9671	2	242	17.8	394
3	0.03237	0.0	2.18	0.0	0.458	6.998	45.8	6.0622	3	222	18.7	394
5	0.02985	0.0	2.18	0.0	0.458	6.430	58.7	6.0622	3	222	18.7	394

```
In [12]: y = df[['MEDV']]  
y.head()
```

```
Out[12]:
```

	MEDV
0	24.0
1	21.6
2	34.7
3	33.4
5	28.7

```
In [13]: x.shape,y.shape
```

```
Out[13]: ((394, 13), (394, 1))
```

```
In [14]: x.info()
```

```

<class 'pandas.core.frame.DataFrame'>
Index: 394 entries, 0 to 504
Data columns (total 13 columns):
#   Column      Non-Null Count  Dtype
---  -
0   CRIM         394 non-null    float64
1   ZN           394 non-null    float64
2   INDUS        394 non-null    float64
3   CHAS         394 non-null    float64
4   NOX          394 non-null    float64
5   RM           394 non-null    float64
6   AGE          394 non-null    float64
7   DIS          394 non-null    float64
8   RAD          394 non-null    int64
9   TAX          394 non-null    int64
10  PTRATIO      394 non-null    float64
11  B            394 non-null    float64
12  LSTAT        394 non-null    float64
dtypes: float64(11), int64(2)
memory usage: 43.1 KB

```

In [15]: `x.describe()`

```

Out[15]:

```

	CRIM	ZN	INDUS	CHAS	NOX	RM	
count	394.000000	394.000000	394.000000	394.000000	394.000000	394.000000	394
mean	3.690136	11.460660	11.000863	0.068528	0.553215	6.280015	68
std	9.202423	23.954082	6.908364	0.252971	0.113112	0.697985	27
min	0.006320	0.000000	0.460000	0.000000	0.389000	3.561000	2
25%	0.081955	0.000000	5.130000	0.000000	0.453000	5.879250	45
50%	0.268880	0.000000	8.560000	0.000000	0.538000	6.201500	77
75%	3.435973	12.500000	18.100000	0.000000	0.624000	6.605500	94
max	88.976200	100.000000	27.740000	1.000000	0.871000	8.780000	100

In [16]: `y.info()`

```

<class 'pandas.core.frame.DataFrame'>
Index: 394 entries, 0 to 504
Data columns (total 1 columns):
#   Column      Non-Null Count  Dtype
---  -
0   MEDV        394 non-null    float64
dtypes: float64(1)
memory usage: 6.2 KB

```

In [17]: `y.describe()`

Out[17]:

MEDV	
count	394.000000
mean	22.359645
std	9.142979
min	5.000000
25%	16.800000
50%	21.050000
75%	25.000000
max	50.000000

In [18]: `x.isnull().sum()`

Out[18]:

CRIM	0
ZN	0
INDUS	0
CHAS	0
NOX	0
RM	0
AGE	0
DIS	0
RAD	0
TAX	0
PTRATIO	0
B	0
LSTAT	0

dtype: int64

In [19]: `y.isnull().sum()`

Out[19]:

MEDV	0
------	---

dtype: int64

In [20]:

```
df = x
df["target"] = y
df.head()
```

Out[20]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	
0	0.00632	18.0	2.31	0.0	0.538	6.575	65.2	4.0900	1	296	15.3	394
1	0.02731	0.0	7.07	0.0	0.469	6.421	78.9	4.9671	2	242	17.8	394
2	0.02729	0.0	7.07	0.0	0.469	7.185	61.1	4.9671	2	242	17.8	394
3	0.03237	0.0	2.18	0.0	0.458	6.998	45.8	6.0622	3	222	18.7	394
5	0.02985	0.0	2.18	0.0	0.458	6.430	58.7	6.0622	3	222	18.7	394

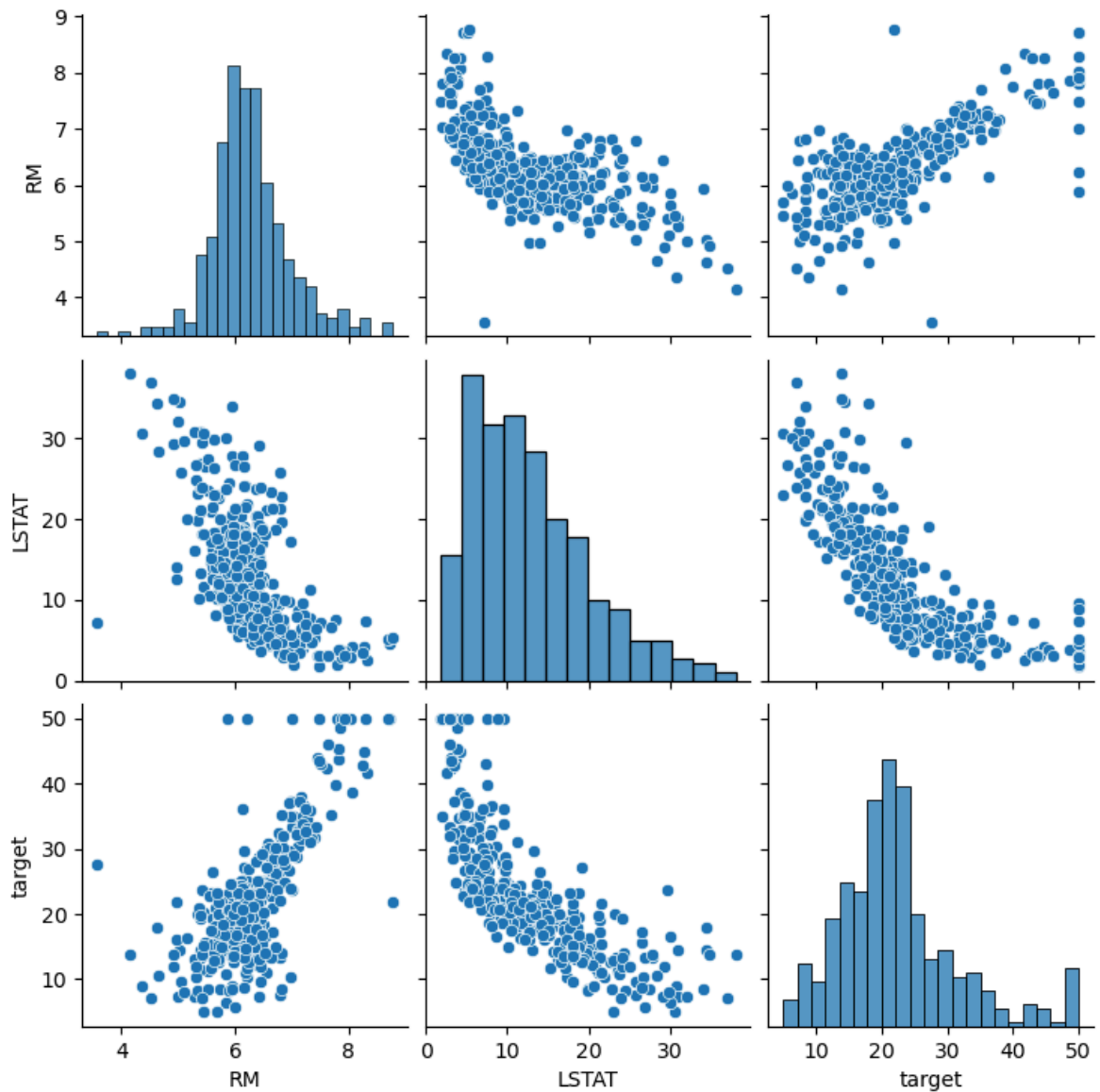
In [21]:

```
plt.figure(figsize=(15,10))
sns.heatmap(df.corr(), annot=True)
plt.show()
```



```
In [23]: df = df[['RM', 'LSTAT', 'target']]
```

```
In [24]: sns.pairplot(df)
plt.show()
```



```
In [26]: x = df[['RM', 'LSTAT']]
         y = df['target']
```

```
In [27]: scaler = StandardScaler()
```

```
In [28]: x = scaler.fit_transform(x)
```

```
In [29]: x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.3,
```

```
In [30]: x_train
```

```
Out[30]: array([[ 1.87899934e-01,  3.08376777e-01],
 [ 3.31351673e-01, -1.02602519e+00],
 [ 2.56632978e+00, -1.17261760e+00],
 [ 2.16590282e-01, -1.05342564e+00],
 [-6.13995292e-01,  2.35656050e+00],
 [-9.90035459e-02, -8.98613094e-01],
 [ 8.23391141e-01, -1.34113038e+00],
 [-9.36761706e-01,  1.38521451e+00],
 [ 3.34220708e-01, -3.39643891e-01],
 [-5.48007491e-01, -1.35785230e-02],
 [ 4.58827113e-02, -2.99913237e-01],
 [ 3.70083643e-01, -6.90369665e-01],
 [ 5.59243331e-02, -2.28672064e-01],
 [-3.28526329e-01,  8.20765213e-01],
 [ 3.08399395e-01,  4.23177018e-03],
 [ 4.23160787e-01, -1.06712587e+00],
 [ 1.29821640e+00, -1.19727801e+00],
 [ 9.32217854e-02, -1.06849589e+00],
 [ 5.00624726e-01,  1.43864539e+00],
 [ 2.25197386e-01,  2.17955289e-01],
 [ 6.41207431e-01, -4.58835853e-01],
 [ 2.89750669e-01, -7.64350883e-01],
 [-5.88173978e-01,  2.19325311e-01],
 [-1.25665909e+00,  6.99928528e-02],
 [-1.01996372e+00,  3.19336958e-01],
 [-3.80168956e-01,  1.24519055e-02],
 [ 1.56340551e-01, -4.31435402e-01],
 [-4.30573674e-02, -9.95884695e-01],
 [-4.74847104e-01, -6.17758470e-01],
 [-9.84100780e-01,  4.39624243e-02],
 [ 1.37137679e+00, -7.99971470e-01],
 [-4.16031891e-01,  2.60425988e-01],
 [ 1.29821640e+00, -1.01095494e+00],
 [-8.13393210e-01,  1.34685387e+00],
 [-2.98401464e-01,  3.71123115e-02],
 [-3.09877603e-01,  2.59055965e-01],
 [-6.15429809e-01,  5.07030048e-01],
 [-9.85535297e-01, -3.68689065e-02],
 [-6.69941470e-01,  9.74207739e-01],
 [ 1.63532799e+00, -8.62992507e-01],
 [-1.97985246e-01,  8.15285123e-01],
 [ 6.74201331e-01, -6.88999643e-01],
 [ 2.83888808e+00, -1.26440912e+00],
 [-4.73412587e-01,  2.04693540e+00],
 [-7.93309966e-01, -2.04011658e-01],
 [-8.24869349e-01, -1.73871162e-01],
 [-1.70279400e+00,  2.31682984e+00],
 [ 2.36549734e+00, -1.24248875e+00],
 [-5.89608496e-01, -4.06774996e-01],
 [-5.62352665e-01,  2.78236281e-01],
 [-1.87493608e+00, -1.76885907e-02],
 [-4.41853204e-01, -3.83484613e-01],
 [-9.72624641e-01,  4.70039439e-01],
 [-2.39586251e-01, -5.35838771e-03],
 [-5.62352665e-01,  1.01503372e-01],
 [-1.07447538e+00,  2.00172466e+00],
 [-6.08257222e-01,  3.16322213e-02],
 [ 5.70916079e-01, -8.90392958e-01],
 [-1.01872581e-01, -7.71200996e-01],
 [-4.47591273e-01, -4.76646147e-01],
```

[-1.25809360e+00, 1.92500339e+00],
[3.29917156e-01, -7.36950432e-01],
[-4.84888726e-01, 5.65941018e-01],
[-1.24824859e-01, 1.19889144e+00],
[1.14615756e+00, -9.68484244e-01],
[6.28296775e-01, -8.50662304e-01],
[2.84892971e+00, -1.18220776e+00],
[-1.40011083e+00, 1.90034299e+00],
[7.80355619e-01, 1.36603419e+00],
[-3.17050190e-01, -5.83507906e-01],
[-6.78548574e-01, 4.46749055e-01],
[1.74989277e-01, -1.98531568e-01],
[-4.16228500e-02, 5.02919980e-01],
[-7.31822327e-02, 3.80987973e-01],
[-1.52080690e-01, 1.56057739e+00],
[-7.64619618e-01, 9.79687829e-01],
[-2.46758838e-01, -4.94456440e-01],
[-1.03307098e-01, 3.28927116e-01],
[-1.01872581e-01, 6.17727175e-02],
[6.42641949e-01, 8.13915100e-01],
[-3.88776060e-01, 6.86228303e-02],
[2.23762869e-01, 4.72779484e-01],
[-5.37965869e-01, -1.80721275e-01],
[2.48149664e-01, -6.21868537e-01],
[-8.85119080e-01, -6.46528943e-01],
[-1.54949724e-01, 8.74196093e-01],
[-1.18636773e+00, 2.44150190e+00],
[-4.26073512e-01, 7.26233657e-01],
[1.22649053e+00, -1.13973706e+00],
[-2.26675594e-01, -9.30123612e-01],
[-6.16864326e-01, -5.47887319e-01],
[2.18331363e+00, -1.48635277e+00],
[1.96507038e-01, -8.28741943e-01],
[6.39772914e-01, -6.46528943e-01],
[1.37691825e-01, 2.57685943e-01],
[1.00414033e+00, -1.05890573e+00],
[2.20893834e-01, 2.23051842e+00],
[-1.33555754e+00, 9.59137491e-01],
[-1.93681694e-01, -8.09561627e-01],
[5.19273452e-01, -4.21845244e-01],
[7.20105888e-01, 1.78389107e+00],
[9.62539328e-01, -4.19105199e-01],
[1.77877973e+00, -7.54760725e-01],
[-5.60918148e-01, 4.02908334e-01],
[-1.02570178e+00, 6.01561604e-01],
[-4.50460308e-01, 2.28915469e-01],
[8.47777936e-01, -9.17793409e-01],
[-4.16228500e-02, 1.94664905e-01],
[1.11870512e-01, -2.12231794e-01],
[4.67630826e-01, -7.39690477e-01],
[-5.73828804e-01, -1.01259967e-01],
[-8.13393210e-01, 2.72756191e-01],
[-7.73226723e-01, 5.21825596e-02],
[2.16590282e-01, 3.59067612e-01],
[-4.44918848e-02, -2.98543214e-01],
[-5.26489730e-01, 4.71409461e-01],
[1.32403771e+00, -4.35545470e-01],
[-9.76928193e-01, 5.12510138e-01],
[-2.16633972e-01, 3.22077003e-01],
[-6.39816605e-01, -5.88987996e-01],

[-8.30607419e-01, 7.64594288e-01],
[-3.95948647e-01, 2.05625086e-01],
[-2.74281911e+00, 2.44698199e+00],
[-1.83620411e+00, 2.63330505e+00],
[4.30136765e-02, -8.24631876e-01],
[-1.21955824e-01, 3.16322213e-02],
[-9.69755606e-01, 1.65894432e-01],
[-5.81001391e-01, 2.27545447e-01],
[9.98402263e-01, -9.76704379e-01],
[2.28803340e+00, -1.34250040e+00],
[-2.56800460e-01, -4.05404974e-01],
[9.71146433e-01, -1.26577914e+00],
[4.08815613e-01, -4.46505650e-01],
[1.47733446e-01, -9.58894086e-01],
[-1.34273013e+00, 1.41398498e+00],
[1.18632404e+00, -5.69807680e-01],
[-5.45335065e-02, -2.77992876e-01],
[-4.94930348e-01, -8.47922259e-01],
[-3.71561851e-01, -3.38273869e-01],
[-2.19503007e-01, -1.76611207e-01],
[-1.36300998e-01, 7.20753566e-01],
[6.85677471e-01, -1.26577914e+00],
[-3.34264399e-01, 6.18001875e-01],
[3.80125265e-01, 1.13861045e+00],
[-1.62122311e-01, -7.20510161e-01],
[-5.06406487e-01, -4.86236305e-01],
[-1.69294898e-01, -7.31470342e-01],
[-1.61385392e+00, 1.00160819e+00],
[-5.86739461e-01, 7.82129881e-02],
[7.37320097e-01, -8.61622485e-01],
[2.96923256e-01, -9.41083793e-01],
[1.62958992e+00, -9.01353139e-01],
[-1.59253277e-01, -9.53413996e-01],
[1.77858312e-01, 1.03311871e+00],
[-8.17893371e-02, 1.23451202e+00],
[-4.54763860e-01, 9.32832363e-02],
[-7.74857849e-02, -3.58824207e-01],
[-7.14411509e-01, 4.34418852e-01],
[7.18671371e-01, -6.98589801e-01],
[-6.88786805e-02, -5.87892674e-02],
[1.95809440e+00, -1.33702031e+00],
[-3.72996369e-01, -3.64304297e-01],
[-1.01852920e+00, 1.13039031e+00],
[-3.97383165e-01, 1.17149099e+00],
[1.26665702e+00, -8.78062755e-01],
[3.58627165e+00, -1.02465517e+00],
[-3.73192978e-02, -3.17723530e-01],
[6.59659548e-02, -6.97494478e-02],
[1.39126342e-01, 1.05229902e+00],
[-5.16448109e-01, 1.20683687e-01],
[1.02996165e+00, -1.34661047e+00],
[1.39002551e+00, -1.09041625e+00],
[5.40791213e-01, 1.15779076e+00],
[-4.93495830e-01, 5.71421108e-01],
[1.25518088e+00, -1.08904623e+00],
[3.50737319e+00, -1.11507666e+00],
[-2.00854281e-01, -3.42383936e-01],
[1.32834126e+00, -6.39678831e-01],
[1.79292829e-01, -6.25978605e-01],
[-8.19131279e-01, 4.13868514e-01],

[-8.22000314e-01, -1.04000012e-01],
[-7.31822327e-02, 1.38219281e-02],
[-5.66656217e-01, -5.45147274e-01],
[-9.79797228e-01, 1.18930128e+00],
[5.09231831e-01, -1.12740686e+00],
[3.95904956e-01, -1.12466682e+00],
[-9.06636841e-01, 1.82334702e-01],
[1.73430969e+00, -1.51238320e+00],
[1.05578296e+00, -1.34387042e+00],
[8.58525894e-03, -5.24596936e-01],
[-1.25665909e+00, 1.53591699e+00],
[-6.97197301e-01, 1.82334702e-01],
[-3.42871503e-01, -6.94479733e-01],
[8.33432762e-01, -1.29317959e+00],
[8.74837158e-02, 6.87873025e-01],
[3.47724833e+00, -1.04794555e+00],
[6.41207431e-01, -1.02465517e+00],
[1.51052497e+00, -6.82149530e-01],
[-8.75077458e-01, 6.61842597e-01],
[4.58827113e-02, -9.06833229e-01],
[-5.91043013e-01, -3.83484613e-01],
[2.25647402e+00, -1.22741851e+00],
[-1.97535230e+00, 2.26339896e+00],
[-3.07275811e+00, 3.45257854e+00],
[2.55322251e-01, -2.31686809e-02],
[7.02891679e-01, -1.08630618e+00],
[3.72952678e-01, -5.64327590e-01],
[2.56756769e-01, 1.53591699e+00],
[-1.62122311e-01, -6.01592899e-02],
[-1.66425864e-01, 1.19067130e+00],
[1.04287230e+00, -9.95884695e-01],
[-4.36115134e-01, 8.55015777e-01],
[-3.93079612e-01, 4.01538311e-01],
[1.09164589e+00, -1.10000641e+00],
[9.96967746e-01, -1.12466682e+00],
[2.20913495e+00, -1.23426862e+00],
[-4.23204478e-01, -4.14995132e-01],
[4.86279552e-01, -7.60240816e-01],
[5.96737392e-01, -7.65720906e-01],
[-2.01050890e-02, -6.67079282e-01],
[-6.69941470e-01, 2.79606304e-01],
[7.20105888e-01, -8.34222033e-01],
[-1.27674233e+00, 1.14683058e+00],
[-2.52620698e+00, 3.31694631e+00],
[7.68879480e-01, -1.20823819e+00],
[-3.68692817e-01, -1.43730666e-01],
[1.64947655e-01, 6.86228303e-02],
[-4.50460308e-01, -3.61564252e-01],
[-9.22416532e-01, 7.63224266e-01],
[1.79292829e-01, 9.04336589e-01],
[1.24513926e+00, -1.26166907e+00],
[-1.36300998e-01, 7.34453792e-01],
[4.51851135e-01, -1.15754736e+00],
[-8.42083558e-01, 6.05671672e-01],
[-7.07238922e-01, -3.46494004e-01],
[-7.08673440e-01, 1.89184815e-01],
[2.07983177e-01, -1.01259967e-01],
[2.59625803e-01, 7.23493612e-01],
[-1.45636284e-03, -1.09480102e-01],
[6.23993222e-01, -3.57454184e-01],

```
[ 7.37320097e-01,  1.16053081e+00],
[-4.26073512e-01,  7.95830107e-02],
[ 3.38524260e-01, -8.78062755e-01],
[ 1.96507038e-01,  8.96116453e-01],
[ 2.51612167e+00, -1.35483060e+00],
[ 4.94886657e-01, -8.38332101e-01],
[ 4.64761791e-01, -1.14932722e+00],
[ 7.63141410e-01, -1.08493616e+00],
[-5.68090735e-01, -6.82149530e-01],
[ 1.96507038e-01, -5.42407229e-01],
[ 3.15375373e-02, -8.28741943e-01],
[ 2.09417695e-01, -7.62980861e-01],
[ 6.42641949e-01, -1.13288695e+00],
[-8.57863249e-01, -3.50604072e-01],
[-3.81603473e-01, -3.06763350e-01],
[-1.18206418e+00, -1.47840733e-01],
[-5.75263322e-01,  6.59102551e-01],
[-5.52311043e-01, -3.02653282e-01],
[-3.90047465e+00, -7.73941041e-01],
[-5.81001391e-01, -5.32817071e-01],
[-1.18493322e+00,  7.24863634e-01],
[-1.24231391e+00,  1.53454696e+00],
[ 7.93266275e-01,  9.48177311e-01],
[-2.33541617e+00,  2.12502669e+00],
[ 2.20913495e+00, -1.26029905e+00],
[ 9.86926124e-01,  6.08411717e-01],
[-2.63973047e-01,  1.03722878e+00],
[ 2.35689024e+00, -1.31646997e+00],
[-2.29544629e-01, -5.05416620e-01],
[ 1.79292829e-01, -2.93063124e-01],
[-1.06730279e+00,  1.48385613e+00],
[ 1.62078620e-01, -1.03835539e+00],
[-3.62954747e-01,  2.13845221e-01],
[-7.50274444e-01,  6.22111942e-01],
[-6.11126257e-01, -1.54690846e-01]]])
```

```
In [31]: model = LinearRegression(n_jobs=-1)
```

```
In [32]: model.fit(x_train, y_train)
```

```
Out[32]: ▼ LinearRegression ⓘ ?
          ► Parameters
```

```
In [33]: y_pred = model.predict(x_test)
```

```
In [34]: y_pred
```

```
Out[34]: array([28.05153405, 19.05560726, 21.8964298 , 30.70586393, 21.34859968,
                25.16011289, 19.01860312, 39.7622007 , 22.26880567, 31.71943521,
                33.0923545 , 31.36949188, 27.05155221, 19.63149959, 19.56840571,
                12.03035107,  6.0446848 , 22.24629821, 35.51353803, 11.92349829,
                33.20540739, 16.062946 ,  6.69714341,  8.46181448, 20.0333215 ,
                23.58410587, 18.21074553, -1.33864446, 14.40707816, 33.02293459,
                21.05793814, 16.96479031, 23.93612557, 25.18438 , 34.69501977,
                24.58527382, 36.18081816, 18.9156103 , 13.67699307, 22.33825957,
                20.28384928,  0.68576323, 12.08458679, 18.93319568, 15.09895215,
                13.86360662, 22.25019216, 17.44529197, 20.85834962, 16.48859937,
                27.89760197, 23.52069026, 23.33916372, 19.77812228, 20.93486803,
                18.34875303, 19.00130858, 19.67956608, 28.44408919, 28.52223631,
                20.51087738, 28.42483918, 23.05933473, 29.55916939, 17.71952528,
                15.65684043, 34.74505179, 10.99911756, 21.66040374, 28.91288194,
                26.98061944, 25.77218399, 28.50671844, 16.76223017, 19.04327191,
                26.72936324, 10.94753174,  5.75649938, 19.61022612, 28.48944784,
                -0.15135154, 29.038539 , 27.04313475, 24.9895618 , 33.30805646,
                4.48160368, 21.17302646, 25.97950247, 18.06509304,  5.12172693,
                11.83108427, 24.15213728, 27.20158028, 31.96290351, 23.04081314,
                25.85545714,  9.25717178, 27.26271366, 27.52935489, 25.94372035,
                14.23749727, 19.61188884, 29.50744569, 16.80791557, 33.58458899,
                26.48817214,  6.93597736, 12.27123936, 19.20099596, 24.78021291,
                24.02167591, 14.96210875, 19.42828091, 28.99375892, 23.8441982 ,
                27.00289709,  8.6256618 , 19.62868475, 21.04241021])
```

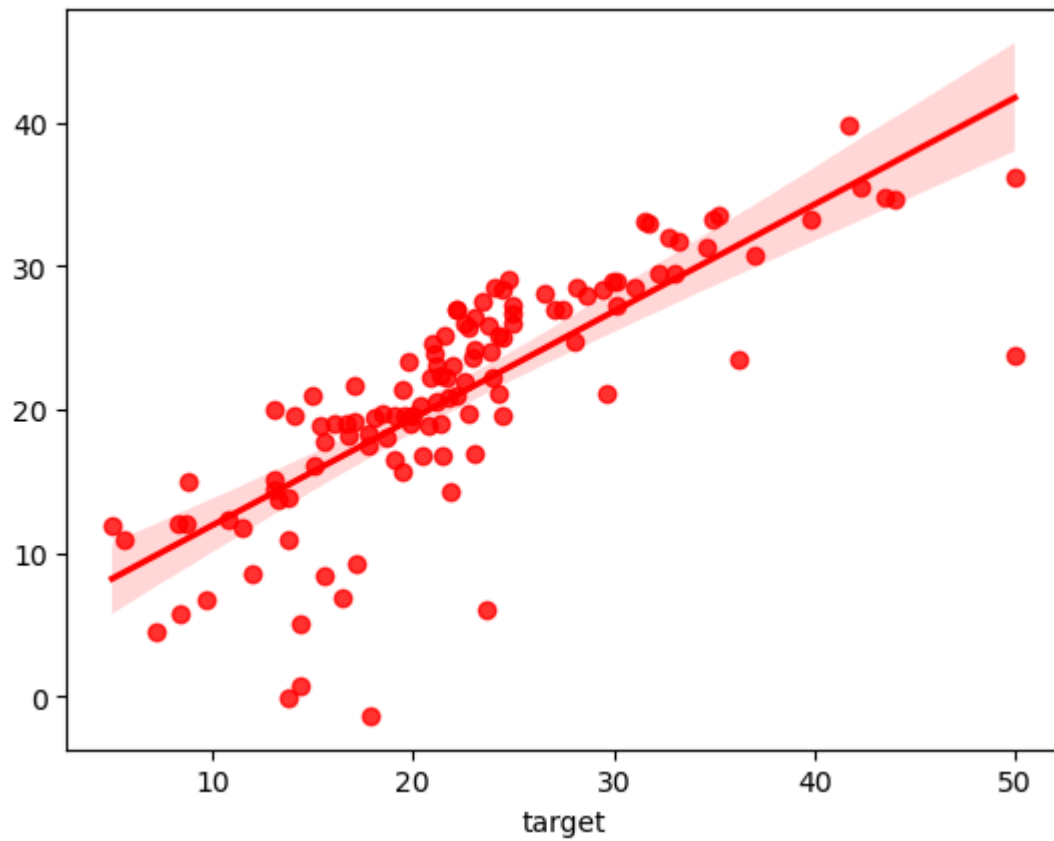
```
In [35]: mean_squared_error(y_test, y_pred)
```

```
Out[35]: 30.551139434909118
```

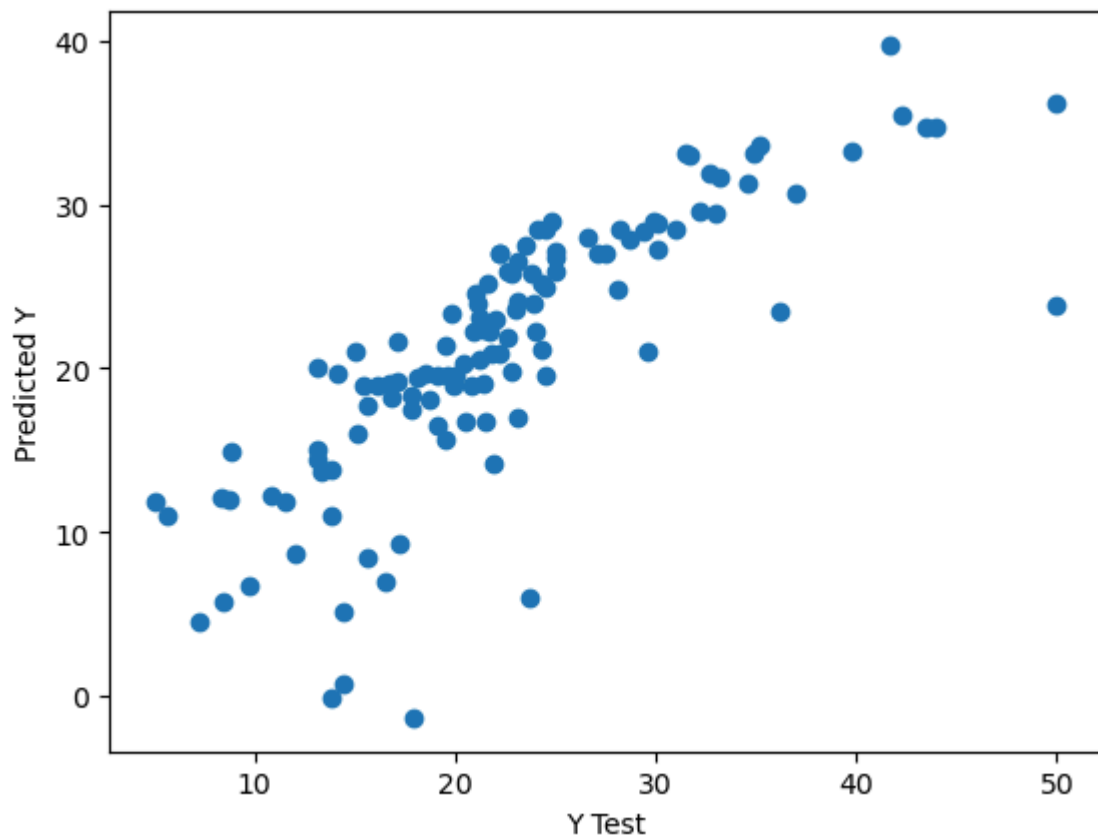
```
In [36]: mean_absolute_error(y_test, y_pred)
```

```
Out[36]: 3.683723499388259
```

```
In [37]: sns.regplot(x=y_test, y=y_pred, color='red')
plt.show()
```



```
In [38]: plt.scatter(y_test, y_pred)
plt.xlabel('Y Test')
plt.ylabel('Predicted Y')
plt.show()
```



```
In [39]: mean_absolute_error(y_test, y_pred)
```

Out[39]: 3.683723499388259

```
In [40]: df1 = pd.DataFrame({'Actual':y_test, 'Predicted':y_pred, 'Variance':y_test - y_pred})
df1.head()
```

Out[40]:

	Actual	Predicted	Variance
85	26.6	28.051534	-1.451534
125	21.4	19.055607	2.344393
174	22.6	21.896430	0.703570
190	37.0	30.705864	6.294136
462	19.5	21.348600	-1.848600

```
In [41]: df.head(15)
```

Out[41]:

	RM	LSTAT	target
0	6.575	4.98	24.0
1	6.421	9.14	21.6
2	7.185	4.03	34.7
3	6.998	2.94	33.4
5	6.430	5.21	28.7
7	6.172	19.15	27.1
8	5.631	29.93	16.5
10	6.377	20.45	15.0
11	6.009	13.27	18.9
12	5.889	15.71	21.7
13	5.949	8.26	20.4
15	5.834	8.47	19.9
16	5.935	6.58	23.1
17	5.990	14.67	17.5
18	5.456	11.69	20.2

```
In [ ]:
```